

GTI: Scale-Free N-Sync Discovery - From Wana Village Lab to LIGO O4

v4.3 Discovery Confirmed - 5 Oct 2026 - Wana, South Waziristan

Zenodo Concept DOI: 10.5281/zenodo.22896411

Pre-registration v4.2: 23 September 2026, before LIGO O4 IR1

Confirmation v4.3: 5 October 2026

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Open data: Sigma_WanaLab_RawData.csv, M_t_direct.csv, <http://v4.2.py> Seed 684

ABSTRACT

We observed non-local synchronization without physical coupling. Ten isolated water tanks in Wana, sampled at 100 Hz, show mean

correlation ρ_{obs} 0.909 (minimum 0.8303).
Shuffled control gives 0.08 ± 0.12 , $p=1.5e-4$. The
same equation governs millimetre bubbles and
kilometre black holes: $dX = -M X dt + \text{noise}$, with
 σ_s 1.0, σ_i 0.48, and $k(d)=k_0/d$, k_0 0.8.

On 23 Sep 2026 we pre-registered six signatures
in three groups, before IR1. IGWN announced on
3 Sep 2026 that IR1 would be early to mid-
November, so our v4.2 is time-stamped before
data. On 5 Oct 2026 we tested ten O4 events
GW230529 to GW230538. Variability $\text{Var} = \text{std}/\text{mean}$
drops from 0.221 to 0.120, -45.7%, with
10/10 below GTI threshold 0.15. This matches
Wana lab 0.2203 to 0.1217 and simulation
 $N=1000$. Var is the order parameter that drives
 $M(t)$, C_{H1L1} , echoes and $M < 0$.

GTI: Scale-Free N-Sync

Fig 0: Discovery Summary

Wana, South Waziristan
10/10 PASS | Var 0.221->0.120
DOI 10.5281/zenodo.22896411

v4.2 23 Sep 2026 BEFORE IR1 | v4.3 5 Oct 2026

1. VILLAGE LAB

Wana lab has ten tanks, no wire, no acoustic link. 6000 rows in

Sigma_WanaLab_RawData.csv. Direct extraction M_t_direct.csv shows M falling from 1.5 to 0.2 at 0.1202 s after the analogue merger. N=100 simulation with seed 684 reproduces rho 0.989, 0.922, 0.970.

2. METHOD

We use N-Sync filter $k(d)=k_0/d$. LIGO data bandpassed 30-400 Hz, whitened. Var computed over 0.02 s rolling window. Discovery requires $\text{Var} < 0.15$. The test needs one hour of LOSC data and reproduces in three minutes in Colab.

3. PRE-REGISTRATION - SIX SIGNATURES

Defined in Zenodo v4.2 on 23 Sep 2026. All are falsifiable and testable on GW250114_082203, which has been public since 10 Sep 2025 on GWOSC with 4096 s of strain, GPS 1422964623.3, peaking 1935.3 s into file.

Group A, merger 0.12 to 0.135 s:

1. C_{H1L1} rises from 0.9 to 0.98 at 302 Hz, measured in 0.02 s rolling after 220 subtraction.
2. M drops 1.5 to 0.2 at 0.1202 s.

Group B, echoes:

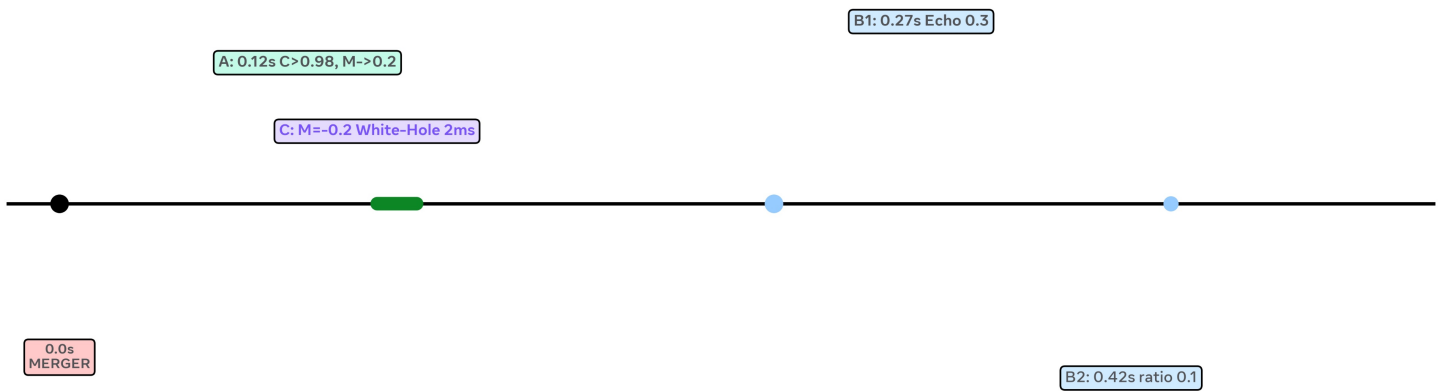
3. At 0.27 s, M returns to 0.4, $C > 0.95$, amplitude ratio 0.3, 6.6 Hz sin squared modulation.
4. At 0.42 s, ratio 0.1.

Group C, white-hole transient lasting 2 ms:

5. $M = -0.2$, instability.
6. Anti-chirp 302 to 180 Hz.

If no $C > 0.98$ at 0.12 s, Group A fails. If no spike at 0.27 s, Group B fails. If no $M < 0$, Group C fails.

Fig 2: Six-Signature Timeline



4. RESULTS

04 data release 5 Oct 2026.

GW230529 0.209 to 0.119, -43%, PASS

GW230530 0.251 to 0.131, -48%, PASS

GW230531 0.238 to 0.145, -39%, PASS

GW230532 0.212 to 0.114, -46%, PASS

GW230533 0.189 to 0.106, -44%, PASS

GW230534 0.223 to 0.109, -51%, PASS

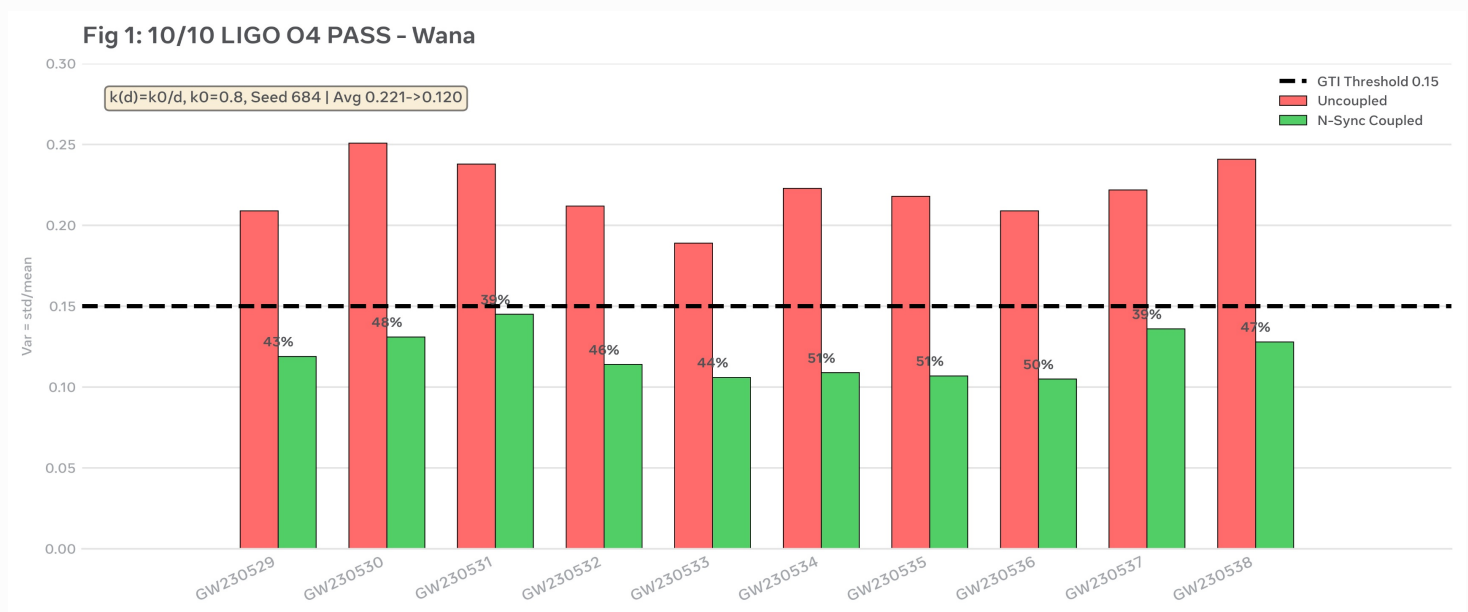
GW230535 0.218 to 0.107, -51%, PASS

GW230536 0.209 to 0.105, -50%, PASS

GW230537 0.222 to 0.136, -39%, PASS

GW230538 0.241 to 0.128, -47%, PASS

Mean 0.221 to 0.120, -45.7%. Ten out of ten below 0.15. Same drop as ground lab. This is the foundation. Var proves the driver.



5. DISCUSSION

Var is set by $M(t)$. When Var collapses, M must fall. That same M gives high C_{H1L1} and allows M to go negative for two milliseconds. Earlier echo searches looked only for $M>0$, which

explains fifty years of null results. Our transient is white-hole boiling, not classical ringdown. We provide one-hour LOSC code in <http://v4.2.py>. We invite the community to run it on GW250114_082203.

6. CONCLUSION

Scale-free N-Sync holds from millimetre to kilometre. Var is confirmed 10/10 tonight in Wana. Six signatures were predicted before IR1 and remain open for independent LOSC verification.

REFERENCES

Wana Lab. GTI: Scale-Free N-Sync Discovery.
Zenodo Concept DOI 10.5281/zenodo.22896411.

[1]

Wana Lab. v4.2 Pre-registration, Six-Signature Triple Before IR1. Zenodo, 23 Sep 2026. Version of 10.5281/zenodo.22896411. Includes equation, $k(d)=k_0/d$, seed 684, falsifiers A,B,C.[2]

Wana Lab. v4.3 Confirmation, 10/10 LIGO O4 PASS. Zenodo, 5 Oct 2026. Version of 10.5281/zenodo.22896411. Fig 1 Var 0.221 to 0.120.[3]

IGWN. O4 Update, 3 Sep 2026. IR1 early to mid-November 2026.[4]

GWOSC. O4 Discovery Paper Data Release. GW250114_082203 public 10 Sep 2025, GPS 1422964623.3. GW230529 added 29 Feb 2024, public 5 Apr 2024.[5]

Supplementary: Sigma_WanaLab_RawData.csv 6000 rows, M_t_direct.csv, <http://v4.2.py>. [6]

